

Fisheries New Zealand

By email: FMsubmissions@mpi.govt.nz

22 July 2026

Review of sustainability measures for Tarakihi for 2026/27

SUBMITTER DETAILS

FULL NAME:	Environmental Defence Society Incorporated
ADDRESS:	PO Box 91736, Victoria Street West, Auckland 1142
CONTACT:	Raewyn Peart
EMAIL:	raewyn@eds.org.nz

Introduction

1. This is a submission on the review of sustainability measures for tarakihi as set out in the Fisheries New Zealand (FNZ) Discussion Paper No: 2026/13 (Review of Sustainability Measures).¹
2. The Environmental Defence Society (**EDS**) is an independent not-for-profit organisation conducting interdisciplinary policy research and litigation. It was established in 1971 with the objective of bringing together the disciplines of law, science and planning to promote better environmental outcomes.
3. EDS has a special interest in the marine environment and recently completed a multi-year project looking at issues within the national oceans management system and options for future reform, which had (among other things) a focus on fisheries matters.²

Summary of Submission

4. Urgent action is needed to ensure sustainability and the long-term viability of a fishery that is on the verge of collapse. There is more than sufficient information on which to act. Any further delay will seriously risk the survival of the fishery.

¹ Fisheries New Zealand (**FNZ**) (2026) *Review of sustainability measures for tarakihi for 2026/27* (FNZ, June 2026, FNZ Discussion Paper No: 2026/13) (**Review of Sustainability Measures**), available [here](#).

² G Severinsen, R Peart, B Rollinson, T Turner and P Parson *The breaking wave: Oceans reform in Aotearoa New Zealand* (EDS, May 2022), available [here](#); R Peart *Pathways to oceans reform* (EDS, September 2025) available [here](#).

5. EDS submits the following points regarding the proposals set out by FNZ in the Review of Sustainability Measures:
 - a. EDS does not support any of the options proposed for catch reductions in the commercial fishery, as they do not adequately address depletion of the tarakihi stocks. Instead, EDS submits the entire fishery should be closed to allow recovery.
 - b. If contrary to the above, recreational fishing is allowed to continue, EDS supports a reduction in the daily bag limit to 2 fish per person. EDS also recommends additional investment into the monitoring, assessment and review of the non-commercial tarakihi fishery.
 - c. EDS supports Option B regarding the closure of key nursery and juvenile aggregation areas, but in addition (and not as an alternative) to full closure of the fishery. These closures should be implemented on a long-term basis, along with further closures of adult spawning aggregation regions.
 - d. EDS submits the minimum legal size (**MLS**) should be increased to at least age of biological maturity to protect immature individuals.
 - e. EDS supports the subdivision of TAR 1 and TAR 7 to better reflect the spatial configuration of the eastern and western tarakihi stocks, and to avoid the use of non-regulatory sub-area catch limits.
 - f. EDS emphasises the need for increased monitoring, and active management, of tarakihi stocks to closely assess the trajectory of biomass and take timely action when needed. This is especially the case in the context of uncertainty, low recruitment and climate change.

Context of the Fishery

6. Tarakihi (*Nemadactylus macropterus*) is a highly valuable inshore commercial fishery in Aotearoa.³ There are thought to be two tarakihi stocks (see Figure 1). The eastern stock ranges from just south of North Cape to along the entire east coast of the country. The western stock extends from North Cape to the lower west coast of the South Island.⁴
7. FNZ describes the management difficulties caused by the overlap of quota management areas (**QMAs**) TAR 1 and 7, with each including part of both the eastern and western stocks, making it hard to manage the stocks separately.⁵ Currently, these areas of overlap are managed with voluntary sub-area catch limits.

³ Ministry for Primary Industries (**MPI**) (2026) *Fisheries Assessment Plenary May 2026 Volume 3 – tarakihi (TAR)* (MPI, May 2026) (**Fisheries Assessment Plenary**) at 1849, available [here](#).

⁴ Review of Sustainability Measures at 1.

⁵ Ibid at [4].

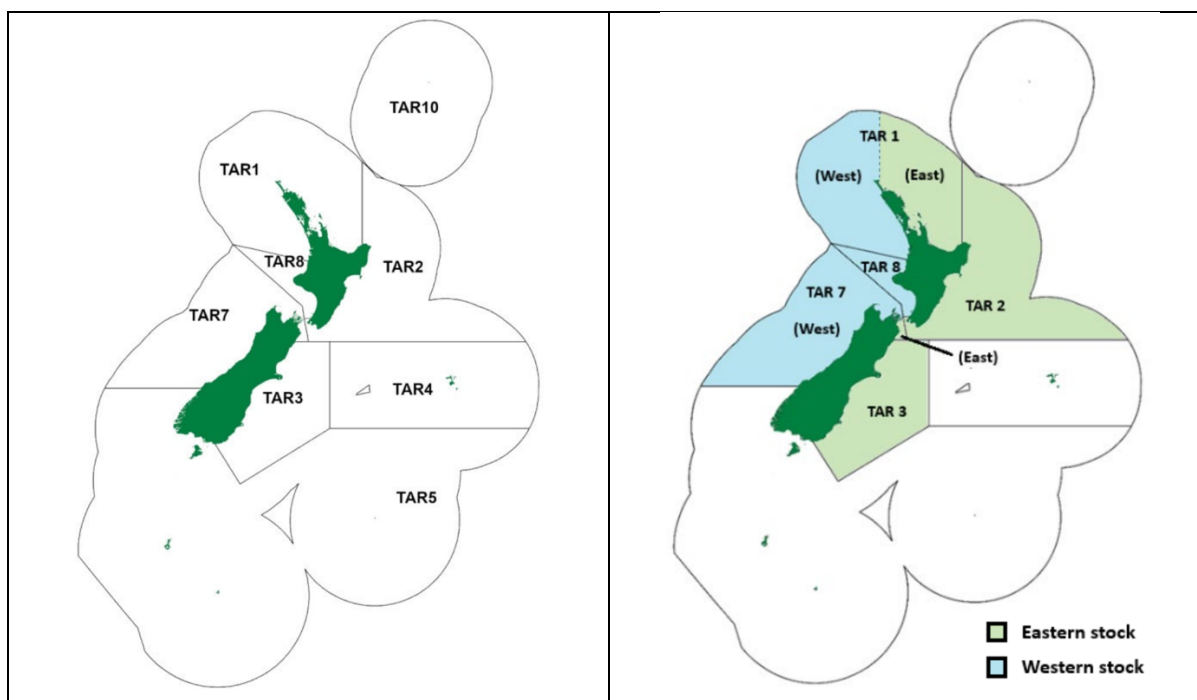


Figure 1. Two maps showing the quota management areas for tarakihi (left)⁶ and the eastern and western stocks (right).⁷

8. Tarakihi spawn in summer and autumn and have a relatively long pelagic larval stage lasting 7 to 12 months.⁸ This can be compared to the short pelagic larval stage of snapper, which can be less than 3 weeks, and means that larval tarakihi are particularly vulnerable to changes in sea conditions.
9. Larvae develop into juveniles the following spring and summer and occupy shallow nursery grounds, generally between 10 and 50 m depth.⁹ Juveniles move into deeper waters up to 200 m at around 25 cm fork length (FL) which is around 3-4 years of age. **While 25 cm FL is the minimum legal size for tarakihi, individuals generally do not reach sexual maturity until 33-37 cm FL (approximately 6-7 years of age).**¹⁰
10. **Accounts of tarakihi nursery grounds in the 1970s characterise them as regions of high benthic biodiversity, with the sea floor dominated by sponge and coral species.**¹¹ Such habitats provide key benefits such as food sources, increased

⁶ Fisheries Assessment Plenary at 1849.

⁷ Review of Sustainability Measures at 1.

⁸ Fisheries Assessment Plenary at 1855-1857.

⁹ M Beentjes *TAR 3 catch sampling in 2009-10 and fishery characterisation (1989-90 to 2009-10)* (Ministry of Fisheries (MF), New Zealand Fisheries Assessment Report 2011/52, November 2011) at 6, available [here](#).

¹⁰ Fisheries Assessment Plenary at 1856-1857; John Annala *The biology and fishery of tarakihi, Nemadactylus macropterus, in New Zealand waters* (Ministry of Agriculture and Fisheries (MAF), Fisheries Research Division Occasional Publication No. 51, 1987) at 8, available [here](#); S Parker and D Fu *Age composition of the commercial tarakihi (Nemadactylus macropterus) catch in quota management area TAR 2 in fishery year 2009-2010* (MF, New Zealand Fisheries Assessment Report 2011/59, December 2011) at 30, available [here](#).

¹¹ C Voren "Nursery grounds of tarakihi (Teleostei: *Cheilodactylidae*) around New Zealand" (1975) 9(2) New Zealand Journal of Marine and Freshwater Research 121 at 134-135; C Voren "Postlarvae and juveniles of the tarakihi (Teleostei: *Cheilodactylidae*) in New Zealand" (1972) 6(4) New Zealand Journal of Marine and Freshwater Research 602 at 615-617.

survival and growth rates, supporting the biomass of many key fish species around Aotearoa.¹² **There are accounts from this period of large quantities of coral being brought up with tarakihi trawls in the Tasman Bay nursery ground.**¹³

11. While the full impact of trawling on these habitats was not known at that time, the risk was recognised.¹⁴ Today, these previously biodiverse inshore regions have been intensely trawled, with regions in Tasman Bay having some of the highest trawl tow counts in the exclusive economic zone between 2008 and 2023.¹⁵ **Recent studies have found that inshore regions have been stripped of the benthic species of sponges and coral, that underpin this biodiversity, as they are vulnerable to trawling.**¹⁶
12. This is of particular concern given that loss and degradation of such habitats can create juvenile bottlenecks, reduce recruitment and make populations increasingly vulnerable to other stressors, such as overfishing and climate change.¹⁷
13. The main nursery grounds of the eastern stock are on the eastern coast of the South Island (TAR 3) around Pegasus Bay and the Canterbury Bight.¹⁸ **Local ecological knowledge of these regions includes both inshore and offshore reefs, which appear to have declined due to fishing activity.**¹⁹ As eastern juveniles mature, they generally move north, with the oldest being found around East Northland and the Bay of Plenty (TAR 1).²⁰ Spawning in the eastern stock occurs in two key regions: TAR 2 from Cape Runaway to East Cape, and TAR 3 from Cape Campbell to Pegasus Bay.
14. The main nursery grounds for the western stock are believed to be around the top of the South Island in Tasman Bay and Golden Bay (TAR 7).²¹ **Local ecological knowledge of these bays suggests that key reefs and biogenic habitat have declined due to the impact of fishing.**²² Some juveniles are also found in TAR 8 around the Taranaki Bight. The key spawning areas for the western stock are thought to be around the Westland coast (TAR 7) and further north around Cape Egmont and Ninety Mile Beach (TAR 1 and TAR 8).²³

¹² See also A Lohrer and others “Benthic Structure and Pelagic Food Sources Determine Post-settlement Snapper (*Chrysophrys auratus*) Abundance” (2018) 5(427) *Frontiers in Marine Science* 1.

¹³ Voren “Nursery grounds of tarakihi”, above n 11, at 134.

¹⁴ *Ibid* at 122.

¹⁵ D J MacGibbon, B A Wood and R Mules *The extent and intensity of seabed contact by trawl gear using fisher-reported and geospatial position reporting data, 1990 to 2023* (FNZ, New Zealand Aquatic Environment and Biodiversity Report No. 372, April 2026) at 17-20, available [here](#).

¹⁶ For example, B Hall and others “Identifying high risk seafloor areas to bottom trawling in Aotearoa New Zealand to support marine spatial management” (2025) 34 *Biodiversity and Conservation* 3481.

¹⁷ See C Brown and others “The assessment of fishery status depends on fish habitats” (2019) 20(1) *Fish and Fisheries* 1.

¹⁸ Review of Sustainability Measures at [34]; Fisheries Assessment Plenary at 1858.

¹⁹ E G Jones and others *Biogenic habitats on New Zealand’s continental shelf. Part 1: Local Ecological Knowledge* (MPI, New Zealand Aquatic Environment and Biodiversity Report No. 174, October 2016) at 39-43, available [here](#).

²⁰ See also J McKenzie and others *Fishery characterisation and age composition of tarakihi in TAR 1, 2, 3, 5, 7, and 8, for 2018-19 and 2019-20* (FNZ, New Zealand Fisheries Assessment Report 2021/79, December 2021), available [here](#).

²¹ Review of Sustainability Measures at [35]; Fisheries Assessment Plenary at 1858.

²² Jones and others, above n 19, at 56-58.

²³ See also McKenzie and others, above n 20.

15. Tarakihi are predominantly caught by bottom trawl, with a small amount coming from longline and set-net fisheries.²⁴ Approximately two-thirds of the tarakihi catch comes from a target trawl fishery, with most of the remainder being bycatch from other inshore trawl fisheries.²⁵

16. Table 1 shows a summary of the total allowable commercial catch (**TACC**), trends in catch levels, and changes in TACC for each QMA. The 1990-91 increases in TACC for TAR 4, 5 and 8 were a result of quota appeals brought when tarakihi was first introduced into the QMS. **This means these fisheries have not had catch limits adjusted as a result of the TACC-setting process since introduction into the QMS 40 years ago.**

QMA	Current TACC	Trend in Catch	Last Change to TACC
TAR 1	978 t	TACC has consistently been under-caught since mid-2010s, landings have been under 1,000 t since 2018-19	2022-23 (3 years ago) Reduced from 1,045 t
TAR 2	1,104 t	Catch peaked in mid-2010s, remaining close to TACC with recent reductions	2022-23 (3 years ago) Reduced from 1,350 t
TAR 3	694 t	Decline since mid-2010s from approximately 1,200 t to 600-700 t	2022-23 (3 years ago) Reduced from 936 t
TAR 4	316 t	Catch has been varied previously, but clear decline since 2018-19 from >100 t to 10-20 t	1990-91 (35 years ago) Increased from 315 t
TAR 5	153 t	Catch peaked at 171 t in 2020-21 and has since declined to under 100 t	1990-91 (35 years ago) Increased from 150 t
TAR 7	994 t	Catch levels have declined from over 1,000 t in the mid-2010s to 600-700 t	2022-23 (3 years ago) Reduced from 1,024 t
TAR 8	225 t	Catch levels have remained relatively steady at 180-250 t since the 2000s	1990-91 (35 years ago) Increased from 208 t
TAR 10	10 t	Catch has remained at 0 t	1986-87 (no change since introduction to QMS)

Table 1. Comparison of current TACC, trend in commercial catch levels, and the last change to the TACC for each of the tarakihi QMAs (as of the 2024-25 fishing year).²⁶

17. These trends in biomass show a tarakihi fishery in decline. The 2018 assessment of the eastern stock noted this trend of depletion. It estimated the stock to be at 17%

²⁴ Review of Sustainability Measures at 1.

²⁵ Fisheries Assessment Plenary at 1854; Review of Sustainability Measures at 1.

²⁶ Fisheries Assessment Plenary at 1849-1854.

of virgin spawning biomass, after increasing slightly from its lowest level in 2014.²⁷ This prompted a 20% reduction in TACC.²⁸

18. Because of depletion, an industry rebuild plan (**IRP**) was introduced into the eastern stock in 2019.²⁹ This included several voluntary measures driven by the industry. Voluntary avoidance areas were implemented in TAR 2 to reduce juvenile mortality, as well as a move-on rule, where vessels would move away and into deeper waters where a tow caught a high proportion of small tarakihi.
19. Additionally, the IRP included a three-phase process towards improving net selectivity to reduce juvenile retention.³⁰ This was completed in 2024, with a range of selectivity trials completed in 2020.³¹
20. **These voluntary measures have clearly not been successful** (see Figure 2). In particular, in the TAR 2 fishery where the voluntary avoidance measures were deployed, the trend has been an increase in juvenile catch, with a major spike during 2025.
21. Sub-area catch limits were implemented in TAR 1 and 7, and while these appear to have been adhered to in TAR 1, in TAR 7 **there has been overcatch by vessels that are not signatories to the voluntary agreement** in 2022-23, 2023-24, and 2024-25.³²
22. **The above highlights the danger of relying on voluntary measures that are not mandatory, are not necessarily backed by science, and cannot be enforced.**
23. In 2019, the 2018 stock assessment was updated, and it estimated stock at 15.9% spawning biomass.³³ This prompted a further 10% reduction in TACC for the eastern stock, which relied heavily on the IRP as an alternative to further regulatory reductions.³⁴

²⁷ A D Langley *Stock assessment of off the east coast of mainland New Zealand* (MPI, New Zealand Fisheries Assessment Report 2018/05, March 2018) at 57, available [here](#).

²⁸ Stuart Nash *Changes to sustainability measures and other management controls for 1 October 2018, and closure of the Kaipara Harbour to the taking of scallops* (MPI, 19 September 2018) at 21, available [here](#).

²⁹ Fisheries Assessment Plenary at 1855; Fisheries Inshore New Zealand, Te Ohu Kaimoana and Southern Inshore Fisheries *Response to the Proposed TAC Changes for the East Coast TAR Fishery (TAR1E, TAR2, TAR3, TAR7E)* (MPI, July 2019), available [here](#).

³⁰ FNZ *The Eastern Tarakihi Management Strategy and Rebuild Plan – Progress Report* (FNZ, Fisheries New Zealand Information Paper No: 2025/09, September 2025) at 11, available [here](#).

³¹ FNZ *The Eastern Tarakihi Management Strategy and Rebuild Plan – Progress Report* (FNZ, Fisheries New Zealand Information Paper No: 2025/06, December 2024) at 10, available [here](#).

³² Review of Sustainability Measures at [110].

³³ A D Langley *An update of the assessment of the eastern stock of tarakihi for 2019* (FNZ, New Zealand Fisheries Assessment Report 2019/41, September 2019) at 10, available [here](#).

³⁴ *Review of Sustainability Measures for tarakihi (TAR 1, 2, 3 and 7 for 2019/20)* (FNZ, FNZ Discussion Paper No: 2019/13, June 2019), available [here](#); See also EDS *Review of Sustainability Measures for tarakihi 2019/20* (EDS, 2019), available [here](#).

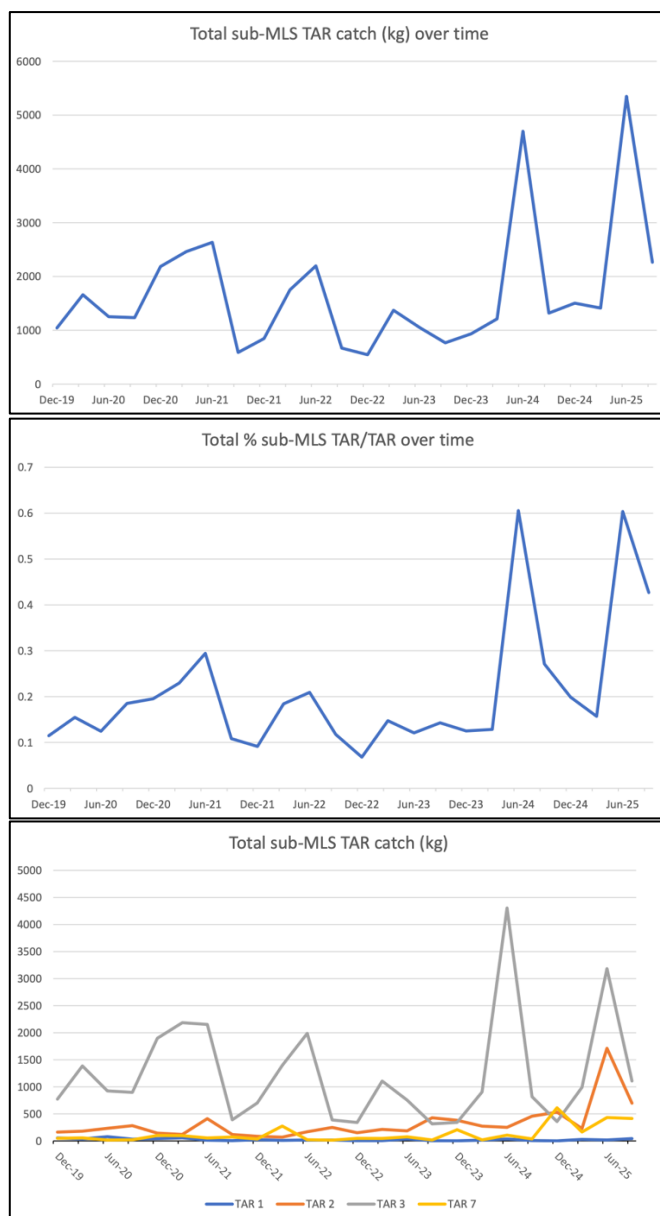


Figure 2. Total sub-MLS catch from the eastern tarakihi stock over time (top), as a percentage of total eastern catch (middle), and by QMA (bottom).³⁵

24. The 2019 decision, particularly due to its reliance on the IRP and the determination of a rebuild period based on economic considerations, was challenged by Forest and Bird.³⁶ The Supreme Court held that the decision was unlawful as the Minister had wrongly considered economic factors in determining the period of rebuild appropriate to the eastern stock.

³⁵ FNZ *The Eastern Tarakihi Management Strategy and Rebuild Plan – Progress Report* (FNZ, Fisheries New Zealand Information Paper No: 2025/09, September 2025) Appendix 2, available [here](#).

³⁶ *Seafood New Zealand v Royal Forest and Bird Protection Society of New Zealand Inc* [2024] NZSC 111.

25. The 2022 assessment of the eastern stock estimated it was around 19% of virgin biomass.³⁷ This suggested it was rebuilding because of reductions in catch. The assessment projected a rebuild to the target reference point of 40% virgin spawning biomass by 2037.³⁸
26. As identified by the 2026 assessment, an error in the definition of how spawning biomass was defined in 2022 (as a number of individuals, not a quantity of biomass) accounted for 53% of the change in biomass noted in that assessment.³⁹ This error also appears to have been made in the 2018 assessment, so was not picked up for 8 years.
27. FNZ suggests that, without this error, eastern tarakihi biomass would have been estimated to be around 12% of virgin spawning biomass, rather than 19%.⁴⁰ This means there has been a continuation of decreasing biomass, rather than stock rebuilding as previously thought.
28. The 2026 stock assessment also included several refinements to the tarakihi stock assessment, including more accurate spatial and sex divisions. **When these refinements are applied to the 2022 assessment, the stock status is estimated at around 8%, rather than 19% as previously assessed.**

2026 Stock Status

Eastern Stock

29. The 2026 assessment estimates the eastern stock at a substantially lower biomass than in previous assessments, at just 7.7% of virgin spawning biomass.⁴¹ Figure 3 demonstrates this stark decline is not a new phenomenon but that **the stock has likely been below the hard limit of 10% spawning stock biomass since before 2010 (ie for more than 16 years).**⁴²

³⁷ A D Langley *A stock assessment of eastern tarakihi for 2021* (FNZ, New Zealand Fisheries Assessment Report 2022/07, March 2022) at 48-50, available [here](#).

³⁸ Review of Sustainability Measures at [37].

³⁹ Fisheries Assessment Plenary at 1882.

⁴⁰ Review of Sustainability Measures at [47].

⁴¹ Fisheries Assessment Plenary at 1870; Review of Sustainability Measures at [42].

⁴² Fisheries Assessment Plenary at 1871.

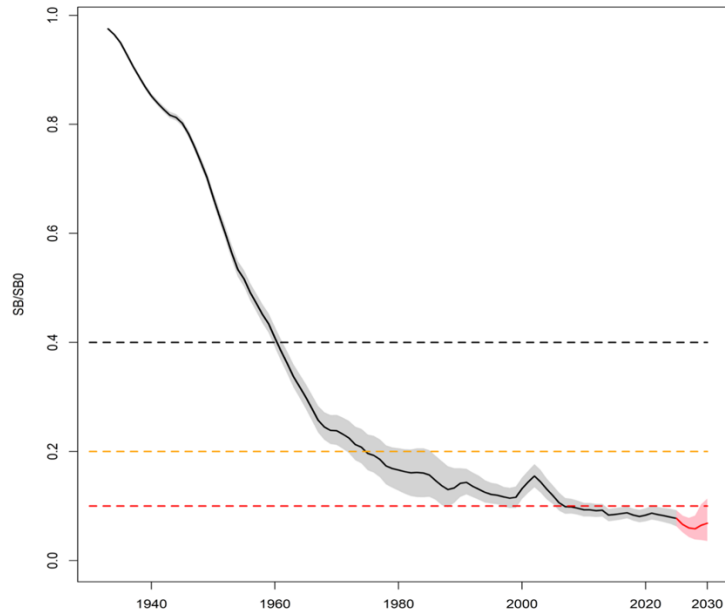


Figure 3. Annual trend in east tarakihi spawning biomass relative to the default 40% SB0 target (black dashed line), 20% SB0 soft limit (orange dashed line), and 10% SB0 hard limit (red dashed line) for the base model.⁴³

30. FNZ additionally states that overfishing is “virtually certain” in the eastern stock.⁴⁴

Fishing mortality of eastern tarakihi has exceeded the overfishing threshold since prior to 1960 and has significantly increased in the decades since. It is currently estimated at 2.6 times the overfishing threshold (see Figure 4 below).⁴⁵

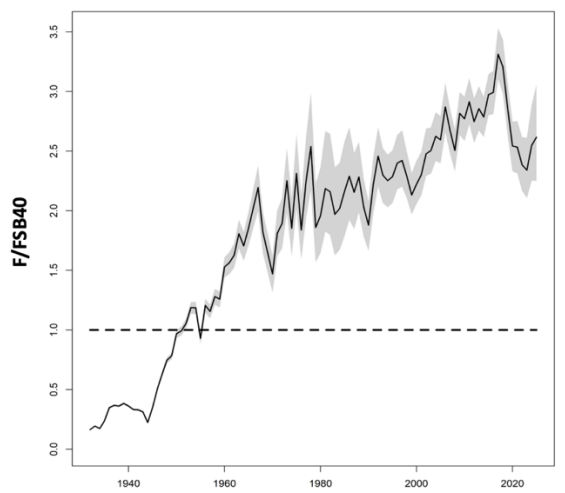


Figure 4. Fishing levels for the eastern stock of tarakihi relative to the overfishing threshold (dotted line). Grey areas show 95% confidence intervals.⁴⁶

⁴³ Review of Sustainability Measures at [46].

⁴⁴ Ibid at [43].

⁴⁵ Ibid.

⁴⁶ Ibid at 9.

31. The stock assessment also found a recent trend of low recruitment.⁴⁷ Given the stocks are characterised by ongoing dispersal and migration between sites of spawning and maturation, **low recruitment threatens the entirety of the eastern stock.**
32. It is important to note that these projections are being measured against *default* management targets and limits, as set out in the Harvest Strategy Standard (**HSS**), rather than those selected as appropriate to the biology of tarakihi which would likely be higher. The Fisheries Assessment Plenary suggested that the lowered tarakihi productivity may mean these default targets are no longer appropriate.⁴⁸
33. FNZ has recognised this, with the adjustments to total allowable catch (**TAC**) proposed under s 13(2A) of the Fisheries Act, rather than s 13(2)(b), on the basis that tarakihi is a stock for which an MSY-producing level is not able to be reliably estimated.⁴⁹
- 34. As a result of these uncertainties, EDS emphasises the need for highly precautionary decision-making. The depleted biomass of the stock means that further reduction in recruitment risks functional extinction.**

Western Stock

- 35. The 2026 stock assessment is the first full assessment of the western tarakihi stock.**⁵⁰ Prior to this, fragmented assessments of parts of the stock were completed. Notably, the TACC of TAR 8 has not been changed since 1986, when it was first introduced into the QMS. Where catch limits have been adjusted, this has been based on stock assessments of the eastern stock. **This is despite recommendations for independent monitoring of the western stock as early as 2001.**⁵¹
36. Given the plummeting stock levels projected in the 2026 stock assessment (see Figure 5), with a 2026 stock estimate of 29% of virgin biomass,⁵² this lack of information has likely contributed to the ongoing depletion of the stock. **It reflects systemic issues of underinvestment in science within the current fisheries management system.**⁵³

⁴⁷ Fisheries Assessment Plenary at 1878.

⁴⁸ Review of Sustainability Measures at [44]; Fisheries Assessment Plenary at 1878.

⁴⁹ Review of Sustainability Measures at [95]; Fisheries Act 1996, s 13.

⁵⁰ Review of Sustainability Measures at [39].

⁵¹ Ibid; S Hanchet and K Field *Review of current and historical data for tarakihi (Namadactylus macropterus) Fishstocks TAR 1, 2, 3 and 7, and recommendations for future monitoring* (MF, New Zealand Fisheries Assessment Report 2001/59, November 2001), at 21, available [here](#).

⁵² Fisheries Assessment Plenary at 1876; Review of Sustainability Measures at [49].

⁵³ See also Environmental Defence Society (**EDS**) *Submission on Proposed Amendments to the Fisheries Act Consultation Document* (EDS, 11 April 2025), available [here](#); *EDS Submission on Fisheries Amendment Bill 2026* (EDS, 6 May 2026), available [here](#).

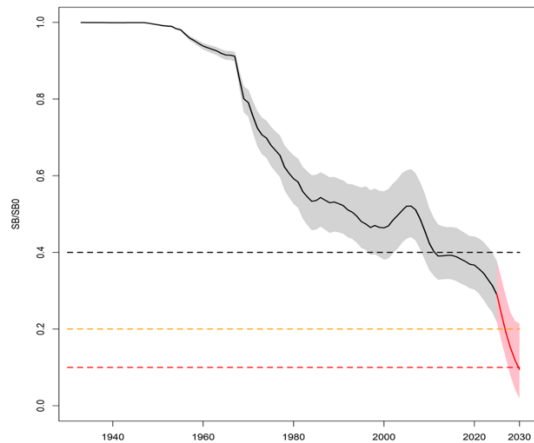


Figure 5. Annual trend in west tarakihi spawning biomass relative to the default 40% SB0 target (black dashed line), 20% SB0 soft limit (orange dashed line), and 10% SB0 hard limit (red dashed line) for the base model.⁵⁴

37. The 2026 assessment suggests that, like the eastern stock, the western stock has faced a significant decline in biomass since the fishery’s establishment in the 1960s and 1970s. There is over a 90% probability that overfishing is currently occurring (see Figure 6).⁵⁵ **Fishing mortality has exceeded the overfishing threshold for decades, and is estimated to have been at 1.6 times the threshold in 2025.** In addition, recruitment is currently below the long-term average.⁵⁶

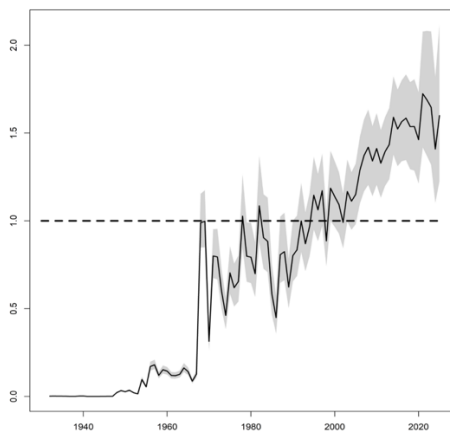


Figure 6. Fishing levels for the western stock of tarakihi relative to the overfishing threshold (dotted line). Grey areas show 95% confidence intervals.⁵⁷

Impacts of climate change

38. For both stocks, recruitment rates have been “well below average” between 2017 and 2022, corresponding with warmer western sea temperatures.⁵⁸ The 2026

⁵⁴ Review of Sustainability Measures at [54].

⁵⁵ Ibid at [51].

⁵⁶ Ibid at [52]; Fisheries Assessment Plenary at 1876-1878.

⁵⁷ Review of Sustainability Measures at 9.

⁵⁸ Fisheries Assessment Plenary at 1878.

assessment notes that, should these conditions persist or become more frequent, the productivity of tarakihi stocks may be reduced.

39. FNZ notes there is “considerable uncertainty” in how recruitment may respond to changes in environmental conditions and fishing within the eastern stock, which “significantly reduces confidence in long-term projections of biomass.”⁵⁹
40. The sea surrounding Aotearoa has warmed at approximately 0.12 to 0.34 °C per decade since the 1980s, with some regions facing even greater rates than this.⁶⁰ The Southern Ocean continues to warm faster than the global average at all depths, due to its absorption of the majority of excess heat from greenhouse gas emissions.⁶¹
41. The east coast of the North Island is also facing the impacts of warming, due to intensification of the East Australian Current, which is causing the region to warm at three to four times the global average.⁶² It is predicted that, by 2050, climate change will result in warming of between 0.8 °C (under an optimistic emissions pathway) and 1.0 °C (under a pessimistic emissions pathway) in the sea adjacent to Aotearoa.⁶³
- 42. The above indicates that the negative impact of warming on tarakihi recruitment, and consequently biomass, may be expected to continue and likely increase in the future.**
- 43. Climate change has not been considered in previous tarakihi assessments, and does not appear to have been integrated in 2026,** including as a measure of uncertainty in stock status or confidence intervals for projections. This is despite reports acknowledging “persistent and predictable biases” in stock assessments caused by climate change.⁶⁴
- 44. It is EDS’s view that the approach taken by FNZ to “wait and see” how the impacts of climate change realise for a species, before accounting for it in decision-making, is unlawful in relation to the information principles specified in section 10 of the Fisheries Act 1996.**
45. As specified in section 10(d) of the Fisheries Act 1996, “the absence of, or any uncertainty in, any information should not be used as a reason for postponing or failing to take any measure to achieve the purpose of this Act.” Additionally, section

⁵⁹ Review of Sustainability Measures at [45].

⁶⁰ P Sutton and M Bowen “Ocean temperature change around New Zealand over the last 36 years” (2019) 53(3) New Zealand Journal of Marine and Freshwater Research 305 at 322.

⁶¹ J Sallée “Southern Ocean Warming” (2018) 31(2) Oceanography 52 at 55-56.

⁶² Sutton and Bowen, above n 60, at 321.

⁶³ C Law and others “Climate change projections for the surface ocean around New Zealand” (2018) 52(3) New Zealand Journal of Marine and Freshwater Research 309.

⁶⁴ See P Neubauer, T A’mar and M Dunn *Climate impacts on fished populations. Part 2: Effects of climate and environmental variability on fishery stock assessment accuracy* (FNZ, New Zealand Fisheries Assessment Report 2023/57, November 2023), available [here](#).

10(c) notes that “decision makers should be cautious when information is uncertain, unreliable, or inadequate”.

Catch Reduction Proposals

46. FNZ proposes three options to reduce fishing pressure on tarakihi through changes to the TACC, recreational harvest allowances and deemed value rates.⁶⁵ FNZ recognises that maintaining the status quo would be inconsistent with its statutory obligations.⁶⁶ So would any delay in taking sufficient action to remedy the situation.
47. **Option 1:** involves a reduction in catch limits of 39% for the eastern stock and 55% for the western stock, resulting in a **46% reduction** across the board.⁶⁷ It would increase deemed value rates and reduce recreational daily bag limits to **ten tarakihi per fisher** (from the current 20). FNZ considers this option to have the least significant impact on utilisation and the “lowest rate and likelihood of recovery”. Importantly, Option 1 would not reduce fishing mortality below the overfishing threshold.⁶⁸
48. **Option 2:** involves a reduction in catch limits of 58% for the eastern stock and 69% for the western stock, resulting in a **61% reduction** across the board.⁶⁹ It would increase deemed value rates and reduce recreational daily bag limits to **five tarakihi per fisher**. Both the rate and likelihood of recovery, and impact on utilisation, would be greater than under Option 1. It would reduce fishing mortality in both stocks to around the overfishing thresholds.⁷⁰
49. **Option 3:** involves a reduction in catch limits of 78% for the eastern stock and 84% for the western stock, resulting in a **78% reduction** across the board.⁷¹ It would increase deemed value rates, although not as much as the other options, on the basis that with a lowered TAC any incidental overcatch is of less risk to the stock.⁷² Recreational daily bag limits would be reduced to **three tarakihi per fisher**.
50. This option would have the greatest impact on utilisation, and FNZ considers it likely to remove target fishing for tarakihi in most QMAs (other than TAR 1 and TAR 2) and require significant avoidance of tarakihi as bycatch.⁷³ In doing so, this option (when compared to Options 1 and 2) would provide the greatest rate and likelihood of rebuild for the fishery. It would also reduce fishing mortality to below the overfishing thresholds for each stock.⁷⁴

⁶⁵ Review of Sustainability Measures at [85].

⁶⁶ Ibid at [81]; Fisheries Act, s 13 (2A)(c)(iii).

⁶⁷ Review of Sustainability Measures at 24.

⁶⁸ Ibid at [152].

⁶⁹ Ibid at 27.

⁷⁰ Ibid at [165].

⁷¹ Ibid at 29.

⁷² Ibid at [173].

⁷³ Ibid at 5.

⁷⁴ Ibid at [175].

51. Notably, none of the proposed options provide projected rebuild periods for either of the stocks.⁷⁵ This is due to significant uncertainty surrounding appropriate management targets, as discussed above.

Commercial Fishery

52. EDS does not consider that any of the proposed options go far enough. **Given the current state of depletion, and projected declines, EDS submits the fishery should be closed until recovery is assured.**

53. In the Review of Sustainability Measures, FNZ has not proposed the closure of either the eastern or western tarakihi stocks. FNZ states that, even for the eastern stock which has a 40-60% probability of being below the hard limit, the proposed reductions would be sufficient to ensure sustainability.⁷⁶ EDS strongly disagrees with this position.

54. The biomass and recruitment trends for both stocks show clear trajectories of collapse. Warming seawater is contributing to this situation through reducing recruitment. We know that seawater will continue to warm in the future, and so the productivity of tarakihi will almost certainly reduce further over time. The most restrictive option proposed (Option 3) will still enable 1,115 tonnes of fish to be harvested each year. EDS submits, given these circumstances, sustainability cannot be *ensured* while continuing such harvest.

55. Allowing such ongoing harvest also fails to provide for utilisation of the tarakihi fishery in the longer term.⁷⁷ The greater a stock's level of depletion, the longer it will take to rebuild, and the longer restrictions will need to be in place. This is particularly the case when spawning biomass is severely depleted, and recruitment limited by a low number of individuals.⁷⁸ A total collapse would preclude any utilisation, with a loss of millions of dollars of foregone revenue in the commercial fishery.

56. Closure of the fishery, as discussed further below, has benefits above and beyond the reduction of direct fishing mortality on tarakihi. For example, requiring vessels to avoid tarakihi will reduce disturbance to spawning aggregations of adults, which are not protected under the proposed nursery and juvenile protections.⁷⁹

⁷⁵ Ibid at [290]-[294].

⁷⁶ Ibid at [26].

⁷⁷ Fisheries Act, s 8.

⁷⁸ C Lundquist and L Botsford "Estimating larval production of a broadcast spawner: The influence of density, aggregation, and the fertilization allee effect" (2011) 68(1) Canadian Journal of Fisheries and Aquatic Sciences 30 at 30-31.

⁷⁹ Review of Sustainability Measures at [226].

Non-Commercial Fishery

57. FNZ considers that current recreational and customary allowances are likely to have a negligible impact on tarakihi biomass.⁸⁰ However, with the potential, for example, of 183 t to be taken in TAR 1 (approximately 19% of the TACC), EDS considers this part of the fishery also needs to be conservatively managed.
58. The exact levels of recreational and customary take are largely unknown. Various forms of recreational surveys have been attempted within the tarakihi fishery, each with their own limitations.⁸¹ The Fisheries Assessment Plenary notes there is “no quantitative information” available for the level of customary catch, although it is thought that some is taken under recreational allowances.⁸²
59. EDS supports the complete closure of the fishery to commercial and recreational fishers. However, if recreational fishing is permitted to continue, the daily bag limit should be reduced to two fish per person. EDS also recommends additional investment into the assessment of both customary and recreational catch to inform a review of the current allowances.

Juvenile Protection Proposals

60. FNZ proposes two options as supporting measures that would protect nursery habitat and juvenile aggregation areas.
61. **Option A:** would restrict trawling, Danish seine and dredging in key tarakihi nursery habitats along the east coast of the South Island (eastern stock) and in Tasman and Golden Bay (western stock).⁸³ Trawling would also be restricted for one juvenile aggregation area for each stock. This is predicted to reduce annual juvenile catch by approximately 1.5 t for the eastern stock, and 140 kg for the western stock.⁸⁴
62. **Option B:** would protect one additional juvenile protection area for each stock, where “methods catching juvenile tarakihi” would be restricted, with a focus on trawling. This is predicted to reduce annual juvenile catch by approximately 2.3 t for the eastern stock, and 230 kg for the western stock.⁸⁵
63. Reporting for the eastern stock IRP shows that sub-MLS catch across TAR 1, 2, 3 and 7 reached approximately 5 t in June in both 2024 and 2025.⁸⁶ Less than half of this would be avoided by either option.

⁸⁰ Ibid at [117], [125].

⁸¹ Fisheries Assessment Plenary at 1855-1856.

⁸² Ibid at 1856.

⁸³ Review of Sustainability Measures at 35.

⁸⁴ Ibid at [213].

⁸⁵ Ibid.

⁸⁶ FNZ, above n5, Appendix 2.

64. The proposed protection measures would be beneficial, particularly in the context of poor recruitment with seawater warming. They would also allow highly trawled benthic ecosystems to recover. However, EDS does not consider they go far enough or defer the need to close the tarakihi fishery.
65. In particular, these measures do not protect spawning adults. FNZ considered spawning area closures in the Review of Sustainability Measures for tarakihi, but has not included them in the current proposals.⁸⁷ This is on the basis that there is “little information” on how fishing activity impacts spawning aggregations, and that such aggregations will be largely avoided under proposed TAC reductions.
66. While there is little information on spawning aggregation protections for tarakihi, there is a significant body of literature on fishing disturbance of spawning, and the use of spawning closures as a common protection mechanism. Van Overzee and Rijnsdorp provide a comprehensive overview of the negative physiological, behavioural, population and ecosystem impacts of such fishing disturbance, as well as of a range of species where spawning closures have been successful.⁸⁸ In Aotearoa, examples include the recent closure of spawning regions in ORH 3B for orange roughy.⁸⁹
67. EDS supports the more extensive protection proposed under Option B and submits they should be complemented with restrictions on trawling, Danish seine and dredging for key adult spawning areas.

Other Measures

Adjustment of MLS

68. FNZ recognises the current MLS for tarakihi (25 cm FL) is smaller than the size at which tarakihi reach sexual maturity (33-37 cm FL), meaning that juveniles are being removed from the population before they can reproduce.⁹⁰ An adjustment to MLS has not been proposed, however, on the basis that there is evidence of low post-release survival of tarakihi from trawls.⁹¹
69. The IRP for the eastern stock has included the development of nets with larger mesh sizes for increased selectivity.⁹² These voluntary measures demonstrate that the

⁸⁷ Review of Sustainability Measures at [224]-[226].

⁸⁸ H van Overzee and A Rijnsdorp “Effects of fishing during the spawning period: implications for sustainable management” (2015) 25(1) Doedrecht 65.

⁸⁹ EDS *Review of proposed spawning area closure to orange roughy fishing in East and South Chatham Rise (ORH 3B) from 1 June 2026* (EDS, 26 March 2026), available [here](#); S Jones *2026 Orange roughy ORH 3B spawning area closure decision* (MPI, 29 May 2026), available [here](#).

⁹⁰ Review of Sustainability Measures at [236].

⁹¹ J McKenzie and others *Estimation of finfish release survival from New Zealand inshore commercial fisheries* (FNZ, New Zealand Fisheries Assessment Report 2024/09, February 2024) at 78-84, available [here](#).

⁹² FNZ, above n 35, 11.

current practice of trawling that results in low juvenile survival is capable of being redesigned – although at present, sub-MLS catch remains substantial.

70. The MLS should act as a regulatory imperative driving industry innovation, rather than a concession to a status quo that threatens the sustainability of the fishery. EDS submits the MLS should be increased to at least the size at maturity.

Subdivision of TAR 1 and TAR 7

71. EDS supports the subdivision of TAR 1 and TAR 7 to reflect the biological distribution of the two tarakihi stocks. This would allow future measures taken in respect of one of the stocks to be regulatory, rather than relying on voluntary sub-area catch limits, as is currently the case.⁹³

Monitoring

72. EDS urges that both of the tarakihi stocks are monitored closely as they (hopefully) rebuild. The failure to regularly assess stocks is an ongoing issue in Aotearoa's fisheries management system.⁹⁴
73. Failure to monitor changes in a stock can lead to TAC setting decisions being inappropriate, and potentially catastrophic. For example, had an assessment of the eastern stock taken place earlier, the error in the 2022 assessment may have been identified and the TAC adjusted earlier.
74. FNZ has suggested the next stock assessment could potentially occur in 2-3 years, rather than 5 years.⁹⁵ EDS submits that, if the fishery is not closed, future assessments of both stocks should be undertaken at least every two years.

Conclusion

75. The tarakihi stocks have been poorly managed and are now near collapse. They require very cautious management if they are to recover, particularly in the context of ongoing seawater warming and habitat loss.
76. Action is urgent and there is ample information on which to act now. EDS submits closure is the only effective option to *ensure* sustainability of the stocks for future generations.

⁹³ Review of Sustainability Measures at [238].

⁹⁴ See also EDS *Submission on Fisheries Amendment Bill 2026* at [64]-[67], available [here](#).

⁹⁵ Review of Sustainability Measures at [62]-[66].